Understanding Negative Numbers

Rina, the curious explorer, is on a journey through the icy lands of Minusville. Temperatures drop below freezing, elevators go underground, and sea creatures swim at negative depths. Can you help Rina understand and solve the chilly number puzzles on her way?

Questions

1. Rina takes a lift in an underground ice hotel. The lift buttons show:

–3, –2, ___, 0, ___, 2, 3 . Which two button numbers are missing?

a) –1 and 1

b) –4 and 4

c) –2 and 2

d) None of the above

2. Rina checks the thermometer in her igloo. It shows –6°C.

The next day, the temperature rises by 4°C. What is the new temperature?

a) –2°C

b) 2°C

c) –10°C

d) None of the above

3. Rina sees a line of animals at different temperatures:

Penguin: **–8°C** Seal: **6°C more than Penguin** Polar Bear: **–5°C** Fox: **8°C more than Bear**. Write the temperature of each animal in the column:

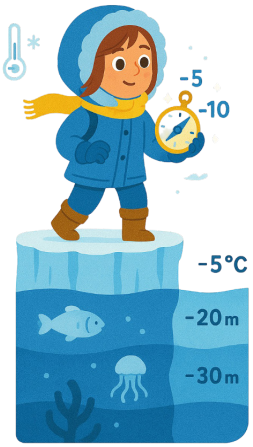
Animal	Temperature
a) Penguin	
b) Seal	
c) Polar Bear	
d) Fox	



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4. Rina's location with her depth:

Location	Depth
a) Top of the glacier	–5 m
b) Ice cave level	–7 m
c) Underwater fish	–3 m
d) Submarine dock	0 m



a) Which location comes in between Underwater fish and Ice cave level..

5. Fill in the blank:

The thermometer shows:

–9, –7, ____, –3, ____, 1, ___, 5

6. Rina says:

"–7°C is warmer than –3°C."

True or False?

7. Rina is on floor **1.** She goes down **3** floors.

Fill in the blank:

She ends up on floor _____.

8. Rina reads two temperature reports:

Mount Snow: –6°C

Valley Breeze: Higher than –6°C but lower than 0°C. List all possible temperatures Valley Breeze could have.

9. Rina and her friends are on different floors of the Ice Tower.

Rina is on floor –2, Lani is on floor 2, and Zed is in between.

Which floors could Zed be on? (Give all possible answers)

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10. Rina dives into an icy lake.

She swims past:

–2 m → –4 m → –6 m → –8 m

Fill in the blanks:

The depths are increasing/decreasing. The next depth she'll reach is _____.

Well done!

Rina has completed her Arctic quest thanks to your help. You've mastered the chilly world of negative numbers and kept your thinking warm all the way!

Well done!
Rina's Arctic Quest



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